

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 14: Automatic Flight Control & Autoland (CAT I/II/III)

Syllabus Source: CAE Oxford (pp. 539-600) | Pass Target: >= 90% AviationExam

1. AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS) HIERARCHY

The AFCS integrates the Autopilot (AP), Flight Director (FD), and Autothrottle / Autothrust (A/T) into an automated flight envelope:

- Flight Director (FD): Computes steering commands displayed as crosshairs or single-cue command bars on the PFD. Indicates where the pilot MUST steer the aircraft manually to intercept and maintain the selected flight path.
- Inner Loop (Stability Augmentation / Fast Loop): Damps aerodynamic oscillations and holds attitude. Senses angular rates from rate gyros/accelerometers and deflects control surfaces (e.g. Yaw Damper to eliminate Dutch Roll). Operates with rapid control surface movements.
- Outer Loop (Guidance / Slow Loop): Tracks navigation paths and targets: Altitude Hold, Heading Select, VOR/LOC, LNAV, VNAV, Glide Path. Displaces control surfaces slowly to guide aircraft along flight trajectory.
- Control Wheel Steering (CWS): A semi-automatic mode where pilot applies manual force on control yoke; autopilot follows pilot input. When yoke pressure is released, autopilot locks onto and maintains the new attitude!

2. ICAO / CS-AWO PRECISION LANDING CATEGORIES

All-Weather Operations (CS-AWO) categorize instrument approaches by Decision Height (DH) and Runway Visual Range (RVR):

Approach Category	Minimum Decision Height (DH)	Minimum Runway Visual Range (RVR)	Autoland System Architecture
CAT I	DH >= 200 ft (60 m)	RVR >= 550 m (or 800 m vis)	Standard single autopilot or manual flight with Flight Director.
CAT II	100 ft <= DH < 200 ft	RVR >= 300 m	Fail-Passive autopilot or dual flight directors. Radio altimeter mandatory.
CAT IIIa	DH < 100 ft or NO DH	RVR >= 175 m	Fail-Passive or Fail-Operational autoland system with automatic rollout/manual rollout.
CAT IIIb	DH < 50 ft or NO DH	50 m <= RVR < 175 m	Fail-Operational autoland system with automatic rollout guidance to runway center.
CAT IIIc	NO DH (Zero DH)	NO RVR (Zero RVR)	Zero-zero landing and taxi guidance. (Not commercially certified due to ground rescue limits).

3. FAIL-PASSIVE VS FAIL-OPERATIONAL AUTOLAND ARCHITECTURE

Transport category aircraft require certified multi-channel redundancy for low-visibility autolandings:

- **Fail-Passive System (Single-Failure Tolerant):** In the event of an autopilot channel failure below alert height, NO OUT-OF-TRIM condition occurs, but the autopilot DISCONNECTS automatically! The pilot must immediately take manual control and execute a missed approach (go-around) if visual reference is not established. Typical for CAT II and CAT IIIa.
- **Fail-Operational System (Dual-Failure Tolerant):** Features at least two independent autopilot channels operating simultaneously (duplex or triplex). If one channel fails below the Alert Height (typically 100-200 ft), the remaining channel continues the approach, flare, touchdown, and rollout AUTOMATICALLY with NO PILOT INTERVENTION! Mandatory for CAT IIIb.

[!] EXAM TRAP (AVIATIONEXAM) - Autoland Flare Mode Trigger & Transition

- Below 50 ft AGL: Sensed by the Radio Altimeter! - The autopilot enters FLARE mode: Pitch attitude increases smoothly to reduce sink rate to ~1-2 ft/sec. - Autothrottle automatically retards thrust levers to IDLE (FMA annunciates 'RETARD' at ~30-20 ft). - At touchdown: Rollout mode tracks the ILS localizer centerline via nosewheel steering and rudder.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Fail-Passive vs Fail-Operational A

Rule 1 (Fail-Passive): Single failure below alert height -> Autopilot disconnects in neutral trim with zero out-of-trim transients. Pilot must see runway and land manually or go around. Rule 2 (Fail-Operational): Single failure below alert height -> Autopilot continues approach, flare, and rollout automatically without pilot intervention (Cat IIIb/IIIc).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In an Automatic Flight Control System (AFCS), what is the difference between an 'inner loop' and an 'outer loop'?

- [A]: Inner loop controls engines, outer loop controls flaps
- [B]: Inner loop provides rate damping and short-term stability augmentation; outer loop provides long-term path guidance (altitude hold, heading, nav tracking)
- [C]: Inner loop is manual, outer loop is automatic
- [D]: Inner loop operates only on the ground

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The inner loop stabilizes the aircraft about its axes using rate gyros (damping Dutch roll and pitch oscillations). The outer loop guides the aircraft along selected navigation paths (holding altitude or tracking localizer).

Q2: What is the defining operational characteristic of a 'Fail-Operational' autoland flight guidance system?

- [A]: Following an autopilot failure, the aircraft automatically executes a go-around
- [B]: Following a single system failure below alert height, the approach, flare, and landing can be completed automatically by the remaining operating channel without pilot intervention
- [C]: It requires manual landing flare

- [D]: It operates on battery power only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Fail-operational systems feature multiple redundant channels. If one channel fails below alert height, the remaining channel continues and lands the aircraft automatically with zero interruption.

Q3: What is the minimum Runway Visual Range (RVR) required for an ICAO Category II (CAT II) precision ILS a

- [A]: 550 m
- [B]: 300 m
- [C]: 175 m
- [D]: 75 m

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. CAT II requires a minimum Decision Height between 100 ft and 200 ft, and an RVR of not less than 300 meters.